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RESEARCH ARTICLE

Operational Stability Enhancement of Pure-Blue Perovskite Light-Emitting Diode With $AgCuGaS_2$ Quantum Dot Solids as Hole Transport Layer

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ABSTRACT

Quasi-two-dimensional (quasi-2D) Ruddlesden-Popper (RP) perovskites characterized by self-assembled multiple-quantum-well structures are promising emitters for high-resolution displays. However, the stabilities of blue perovskite light-emitting diodes (PeLED) are major hurdles toward full-color applications. Here, we introduce wide-gap, high-mobility silver-copper-gallium-disulfide ($AgCuGaS_2$, ACGS) quantum dot (QD) solids as novel hole transport layer (HTL) to address the bottlenecks of the short lifetime of blue PeLEDs through controllable charge transport. ACGS QD solids provide enhanced hole transport through adaptable band structure and high hole mobility of 0.191 $cm^2 V^{-1}s^{-1}$. In addition, Zn^{2+} attached to surface of the ACGS QD, stabilize the halide anions through ionic interaction, thus suppressing ion segregation and lattice defect formation in quasi-2D perovskite. The resulting pure-blue PeLEDs (471 nm) based on ACGS QD solids achieve a maximum EQE of 6.92% and a record operating lifetime with T_{50} (luminance drop to half of the initial value) of 78 min at 100 cd/m^2, significantly higher than controlled devices based on organic HTL materials (18 min).

1 | Introduction

From the demonstration of first room-temperature electroluminescent perovskite light-emitting diodes (PeLEDs) in 2014, the past full decade has witnessed sustained development of luminescent perovskite materials, as well as remarkable advances in highly-efficient and stable devices, shining as promising alternatives to future lighting and displays [1–7]. Despite significantly improved efficiency and luminance properties in red, green, and near-infrared perovskite emitters, the major research and

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development obstacles relate to the inferior operational stability and device longevity of blue pixels.

Quasi-two-dimensional (quasi-2D) perovskites with Ruddlesden-Popper (RP) phase are constructed by layered 2D perovskite bonded through large organic spacer cations with low conductivity, such as butylammonium (BA), phenylethylammonium (PEA), etc. [8]. The advantages of quasi-2D perovskite include larger exciton binding energy (hundreds of meV) and lower ion migration mobility compared to 3D counterparts, thus having gained great success in blue PeLEDs over the past five years [9, 10]. State-of-the-art blue PeLEDs based on quasi-2D perovskite have reached external quantum efficiencies (EQE) close to their red and green counterparts, over 25% for sky-blue (490 nm), 15% for pure (475 nm), and deep-blue (465 nm) devices. However, blue emitters and LEDs still suffer from extremely short operational lifetime, with T_{50} (time to 50% of initial luminance) of only a few hours for sky-blue devices, and even a few minutes for deep-blue ones [11–15]. In quasi-2D perovskite, a deeper blue emission required by displays (less than 470 nm) is usually achieved through compositional engineering of halogen anions (Br^-, Cl^-), accompanied by incorporating a functionalized large organic spacer to spatially-confine excitons more strongly [16]. However, the long-term stabilities of mixed-halide quasi-2D perovskite are severely restricted by tremendous deep-level defect formation during operation, thus leading to exciton quenching in perovskite emitters [17–19]. For example, Yang et al. have revealed high mobility of Cl^- under the influence of an electric field, and the chlorine oxidation play significant role in device failure [20].

It is widely acknowledged that charge transport in thin-film optoelectronic devices exerts decisive effects on long-term stabilities [21]. In quasi-2D PeLEDs, charge accumulations at either hole or electron transport interface will induce dramatic material degradation and numerous deep defect states [22, 23]. On one hand, imbalanced charge transport will introduce localized electric field within the emissive layer to drive the migration of halogen ions, leading to lattice distortion and the formation of vacancy defects [24]. On the other hand, a space charge layer due to mobile ions gradually forms at the interface, screening the external bias and reducing the potential drop across the emissive layer [25]. To operate under higher luminance, a larger external voltage is necessary, which would further promote ion migration and generate more deep-level vacancy defects. Apart from ion migration, it was recently affirmed that the existence of metallic lead at the perovskite/ETL interface through in situ grazing- incidence wide-angle X-ray scattering (GIWAX) analysis, indicating electrochemical reactions at the surface of the charge transport layer due to charge accumulations [26]. Therefore, it is evident that charge transport imbalance are critical underlying issue toward device degradation in quasi-2D blue PeLEDs through accelerated defect formation. Though intense effort has been spent on additive engineering to stabilize or passivate ionic defects in quasi-2D perovskite emitters via chemical bonding effect, interfacial charge transport, and their effects on device operational stabilities have received less attention.

In terms of charge transport material, conventional organic polymer including poly(9,9-dioctyl-fluorene-co-N-(4-(3-methylpropyl))diphenylamine (TFB), and Poly(N-vinyl carbazole) (PVK), have been widely implemented in solution-processed blue PeLEDs as HTL. However, these organic polymers suffer from lower carrier mobility compared to electrons, thus leading to charge imbalance [21, 27]. From our preliminary results, it was affirmed that the hole injection rate is much slower than the electron, due to low hole mobility. We fabricated a double-emitter device embedded with a top layer of red InP/ZnSe/ZnS quantum dots and a bottom layer of blue quasi-2D perovskite (Figure S1). From the electroluminescence (EL) spectrum, the red emission from this double-emitter device is negligible compared to blue at the turn-on voltage point. This reveals that the recombination zone is mainly located at the HTL-blue perovskite interface. Inorganic metal oxide nanoparticles, including nickel oxide (NiO_x) and molybdenum oxide (MoO_x) has been developed as highly-conductive and stable HTL in PeLEDs [28]. Nevertheless, EL efficiencies of these PeLEDs are dramatically limited by degradations of perovskite nanocrystals due to chemical interactions between under-coordinated metal cations and perovskites [29, 30].

Motivated by dramatic success in applying NiO/self-assembled molecules (SAM) for selective hole transport in perovskite photovoltaics, we seek to realize highly-stable pure-blue PeLEDs by implementing a layer of wide-gap semiconductor quantum dot (QD) solids with tens-of-nanometer thickness as novel HTL [31]. Different from SAM, which are single layer of organic molecules, the developed novel HTL herein is constructed by mono-dispersed colloidal QDs, which are self-organized into solid thin-films by solution process. In particular, the chalcopyrite I-III-VI_2 family ($CuInSe_2$, $CuInGaS_2$, etc.) is selected due to its intrinsic p-type semiconducting properties, featured by high hole mobility and tunable bandgap (E_g) [32]. Compared to NiO_x, of which the hole conductivity is mainly determined by the concentration of Ni^{3+}, chalcopyrite QDs could be delicately engineered via stoichiometry, cation doping, QD size, and surface chemistry [33, 34]. Herein, we show that strong electrical coupling could be realized in $CuGaS_2$ (CGS) QD solids by atomistic ligand exchange, leading to exceptionally-high hole concentration (5.81×10^{15} cm^{-3}) and mobility (0.546 cm^2 V^{-1} s^{-1}) in the linear working scheme. Most importantly, the Z-type surface ligand embedding Lewis acid (Zn^{2+}) of ACGS QDs could stabilize halide anions, suppressing ionic migration, thus preventing halide vacancy defect formation. With balanced charge transport and reduced defect density, quasi-2D pure-blue PeLEDs prepared in this work demonstrate improved luminescent performances with peak external quantum efficiencies (EQE) of 6.92% at 471 nm, and exceptional operating lifetime of 78 min (T_{50}@100 nit), significantly higher compared to devices based on NiO_x (18 min).

2 | Results and Discussions

2.1 | Synthesis and Adaptable Band Structure of I-III-VI_2 Semiconductor Quantum Dot

CGS core QDs were synthesized and assembled into QD solids as HTL in blue PeLEDs based on quasi-2D perovskite, as depicted in Figure 1. Figure S2 shows the synthetic scheme and material characterizations of CGS QDs prepared in this work. From high-resolution transmission electron microscopy (HR-TEM) images,

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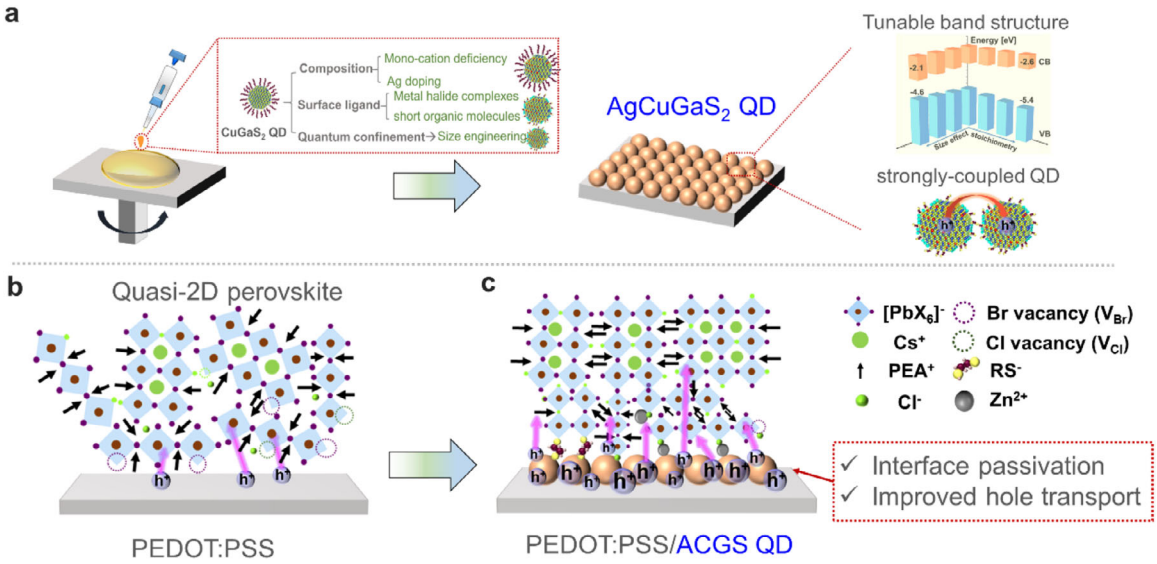

FIGURE 1 | Schematics of proposed blue PeLEDs with novel HTL in this work. (a) Fabrications of ACGS QDs with delicate manipulation on chemical composition, surface ligand, and size during synthesis, to achieve tunable band structure and enhanced electrical coupling in QD solids, which would facilitate improved hole transport in PeLEDs. (b,c) illustration of defect passivation of quasi-2D perovskite films with (c) and without incorporating ACGS QD solids (b), through stabilizing halide anions. The enhanced hole transport from ACGS QD solids into quasi-2D perovskite is highlighted in pink solid lines. The octahedral of $[PbX_6]^-$, Cs^+, PEA^+, as well as functional surface ligand (RS^-, Zn^{2+} ions), are pinpointed at the right-hand side. Ionic migration through halide vacancy (V_{Cl}, V_{Br}), denoted as open dashed circles, is largely suppressed due to stabilization of Cl^- and Br^- interstitials through binding to Zn^{2+} cations.

CGS QDs exhibit uniform spherical shape with an average size of 4.25 nm (Figure S2b), comparable to the Bohr exciton radius (~4 nm) of chalcopyrite nanocrystals [35]. X-ray diffraction (XRD) patterns of CGS QDs present dominant (112) planes attributed to the chalcopyrite phase (Figure S2d). Fast Fourier Transform (FFT) analysis reveals single-crystal character of CGS QDs, consistent with chalcopyrite structure in the [1-10] direction.

Recent reports reveal that doping with silver (Ag) into chalcopyrite prompts uniform nucleation and reduced deep-level lattice defects [36]. Moreover, due to orbital overlap between Ag 3*d* electrons and Cu 4*d* electrons, the valence band maximum (VBM) level is lowered, which would reduce the hole transport barrier with the quasi-2D perovskite. Herein, Ag doping was conducted by incorporating an appropriate amount of silver iodide (AgI) into metallic precursors during synthesis. The elemental mapping of ACGS QDs dispersions was measured by STEM-EDX as displayed in Figure S2c, indicating successful doping with Ag. The purified $AgCuGaS_2$ (ACGS) QDs exhibit a similar spherical shape without affecting size distribution (Figure S3). To investigate the band structure of ACGS QD solids, we performed UV–vis absorption measurement to derive the bandgap energy (E_g) of ACGS QD solids. In particular, E_g increases from 2.17 to 2.78 eV as [Cu]/[Ga] (molar ratio in the precursor) increases from 1:1 to 1:8 in CGS; and increases conspicuously from 2.58 to 2.74 eV as AAC increases from 0% to 10% in ACGS (Figure S2e,f). It is important to note that ACGS QDs cannot be synthesized under stoichiometric precursor conditions (([Ag]+[Cu])/[Ga], ACG = 1.0), due to a large amount of crystalline Ag_xS/Cu_xS QDs preformed, obstructing the Ga diffusion (Figure S4). Detailed investigations into the nucleation and growth process of QD affirmed the formation of crystalline Cu_2S/Ag_2S during early stages of synthesis (Figure S5 and Table S1).

The work function (WF) and energy level of valence band maximum (VBM) were detected from ultraviolet spectroscopy (UPS). The experimental UPS spectrum, as well as the estimated WF and VBM, were shown in Figure S6. First, Ag doping lowers the VBM energy level due to reduced hole concentration. From Figure S6b, as the ACG ratio is fixed at 1:2 and the AAC ratio increases from 0% to 10%, the VBM deepens from −4.53 to −4.93 eV. Second, from Figure S6c, increased Cu deficiency leads to remarkably lowered Fermi-level (E_F) position and the VBM. For instance, as the AAC ratio is 0, and the [Cu]/[Ga] ratio decreases from 1:2 to 1:8, the VBM is lowered from −4.93 to −5.37 eV. Then, the quantum confinement effects also affect the band structure; when we synthesized larger ACGS QDs (average size ~5.37 nm) using chloride precursors (CuCl, $GaCl_3$), remaining other parameters unchanged (ACG = 1:2, AAC = 10%), the VBM is upshifted to −4.50 eV compared to −4.93 eV for smaller QDs (Figure S6d,e). In terms of surface ligand effects, we didn't affirm direct evidence about band-tuning effects through ligand engineering in this work. However, in lead sulfide QDs (similar crystal structure to I-III-VI_2 material), the valence band level could be feasibly modulated via ligand types [37]. Thus, it is notable that the band structure of chalcopyrite QD solids could be feasibly manipulated through composition, ligand engineering, and quantum confinement effects (Figure S6f). Next, the microscopic surface contact potential of quasi-2D perovskite films, as well as various hole transport material were estimated through Kelvin probe microcopy (KPFM) as shown in Figure S7. Here, the ACGS QDs were synthesized with AAC ratio of 10% and ACG ratio of 1:8. The contact potential difference (CPD), representing the WF differences between sample surface and probe, was measured to be −223, 14.7, 178 mV for quasi-2D perovskite, ACGS QD solids and poly (3,4-ethylenedioxythiophene)-poly (styrenesulfonate) (PEDOT:PSS), respectively, corresponding to a E_F position of

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−4.68, −4.92, and 5.08 eV Therefore, the E_F of ACGS QD lies between PEDOT:PSS and perovskite, signifying its critical role as an energy cascade to assist hole injection from PEDOT:PSS into the perovskite layer.

2.2 | Functional Ligand Exchange and Improved Hole Mobility in ACGS QD Solids

Prior to using ACGS QD solids as HTL in PeLEDs, we developed hybrid ligand exchange technology to replace the long-bulky ligand (RNH_2, RS^-) on the surface of QDs with short, atomistic ligand [38], as illustrated in Figure 2a.

At the beginning, $ZnCl_2$ was intentionally selected as a solution-phase ligand treatment due to the similar lattice constant between chalcopyrite and zinc-blende ZnS crystals [39]. $ZnCl_2$ dissolved in ethanol was added to roughly-purified QD solutions to initiate ligand interactions (see Experiment Section for more detail). The schematic of surface ligand exchange is depicted in Figure 2a.

Before ligand exchange, pristine ACGS QDs exhibit distinguished lattice distortions and stacking faults due to steric hindrances, which could be effectively recovered with improved crystallinity characterized by an obvious decrement of inter-dot spacing and uniform distribution after $ZnCl_2$ treatment, promoting stronger electrical coupling among dots (Figure 2b; Figure S8c). The crystallography characterizations by XRD shown in Figure S8d present trivial variations in the diffraction pattern after ligand exchange, indicating the process is a surface-limited interaction, instead of alloying or doping into the core. To understand microscopic origins of Cl^- passivation on QDs, we used density functional theory (DFT) to calculate the binding energies (E_b) of bulky ligand (RNH_2, RS^-) and Cl^- ions on the CGS surface. In Figure 2c, DFT calculations unveil that Cl^- could bind to surface Cu^+ and Ga^{3+} of CGS QDs with stronger Lewis acidity, reconstructing ordered cubic structure from distorted crystal surface due to large steric hindrances of the aliphatic ligand. Furthermore, from density of states (DOS) calculations, the mid-gap trap states are remarkably suppressed, and E_b is significantly improved after $ZnCl_2$ treatment (Figure 2d; Figure S9). X-ray photoelectron spectroscopy (XPS) studies were conducted to characterize the chemical environment of the element in ACGS QDs on ligand exchange. As expected, Zn 2*p* and Cl 2*p* were distinctly identified in $ZnCl_2$-treated ACGS QDs (Figure S10). In addition, both Cu 2*p 1/2* and Ga 3*d* emission peaks, as well as corresponded auger emissions (Cu *LMM*, Ga *LMM*), slightly shift to a higher binding energy regime, due to bonding with electron-rich Cl^- ions (Figure S10h,i). The metal nitride peaks, evidenced from resolved Ga 3*d* (25.68 eV) and N 1*s* (396.8 eV) core-level emission, are efficiently eliminated in ACGS QDs after $ZnCl_2$ treatment, which can also be confirmed by quantitative analysis of atomic composition through XPS fine scan (Table S2). Elemental analysis from XPS spectra indicates reduced atomic concentration of nitrogen (Table S3). These results clearly indicate partial exchange of surface amines (RNH_2) by Cl^- ions. Nevertheless, $ZnCl_2$ cannot sufficiently remove the residue thiolate ligand from 1-Dodecanethiol (1-DDT) used as surfactant, which can be recognized from thermogravimetric analysis (TGA) and Fourier transform infrared spectroscopy (FTIR) analysis, in which the methyl and methylene vibrations are still present inACGS QDs after $ZnCl_2$ treatment (Figure S11). To fully strip the un-exchanged alkyl ligand on ACGS QDs, we explored a solid-phase ligand exchange process to produce homogenous QD solids using 1,2-ethanedithiol (EDT) containing a double RS^- group [40]. Likewise, DFT simulations show that EDT exhibits further enhanced coordination with surface gallium compared to Cl^-, leading to increased E_b from 3.193 to 9.747 eV, and fully-passivated defect states. High-resolution XPS spectra suggest trivial influences of EDT treatment on chemical bonding and atomic concentrations of elements in ACGS QDs. Meanwhile, the Cu and Ga concentration is declined after $ZnCl_2$ treatment, as summarized in Table S3. With observations of QD size decrement and blueshifted absorption spectrum of ACGSQDs prone to $ZnCl_2$ treatment (Figures S8b and S11c), it is reasonable to address that the surface of ACGS QD is "cleaned" via desorption of the surface metallic complexes, which failed to crystallize into chalcopyrite phase, through surface ligand exchange, leading to reduced surface defect [41]. As $ZnCl_2$ removes ligand excessively, ACGS QDs tend to aggregate in the solution, detected by dynamic light-scattering (DLS) measurement (Figure S11d). The influences of ligand density and film washing with EDT on surface morphology of QD solids were monitored through scanning electron microscopy (SEM) and atomic force microscopy (AFM) (Figure S12). As the ligand-to-QD ratio exceeds 100:1, the deposited films become rough and fuzzy, which is detrimental for photon out-coupling. The optimized ligand-to-QD ratio is limited to 100:1 for $ZnCl_2$ treatment (see Experiment Section). The stoichiometric elemental composition is also achieved for ACGS QDs treated by $ZnCl_2$ with a ligand-to-QD ratio of 100:1 (Table S4). Consequent solid-phase exchange is optimized to be a single coating with EDT (1 vol.% in acetonitrile). The as-synthesized and exchanged ACGS QDs by $ZnCl_2$ and EDT are named as ACGS-pristine, ACGS-$ZnCl_2$, and ACGS-$ZnCl_2$-EDT, respectively.

Next, we built up a thin-film transistor model with a bottom-gate-top-contact structure using ACGS QD solids as the transport channel (QD-TFT) to quantitatively assess the carrier mobility and concentration. Figure 2e depicts the device structure, and the corresponding photograph of the SiO_2 substrate, as well as the SiO_2/ACGS QDs under the optical microscope, is shown in Figure S13a. By altering the concentration of QD dispersion, the thickness of QD solids can be practically controlled (Figure S13b, supporting information). For TFT analysis, the ACGS QD were synthesized with a common ACG ratio of 1:8, with a varied AAC ratio from 10% to 40%.

Hole-dominant transport can be surely confirmed from transfer characteristics (I_{DS}-V_{GS}) in QD-TFT devices (Figure 2f). I_{DS} collected from QD-TFT based on ACGS-pristine is significantly increased from 0.1 to 20 nA after $ZnCl_2$ treatment, and further elevated to hundreds of nano-ampere (nA) after EDT treatment. Corresponding output characteristics (I_{DS}-V_{DS}) of QD-TFT based on ACGS-$ZnCl_2$-EDT QD solids are depicted in Figure 2g, signifying a saturation region while V_{DS} exceeds -2 V. The influences of chemical composition on transfer characteristics of QD-TFT based on ACGS QDs are reflected in Figure S14. I_{DS} increases as the [Cu]/[Ga] ratio increases from 1:8 to 1:2. This can be explained by the formation of large amounts of nano-crystalline Cu_xS phase in CGS QDs as the [Cu]/[Ga] ratio increases. Contrarily, for ACGS QDs, I_{DS} decreases as AAC ratio increases, which can be explained

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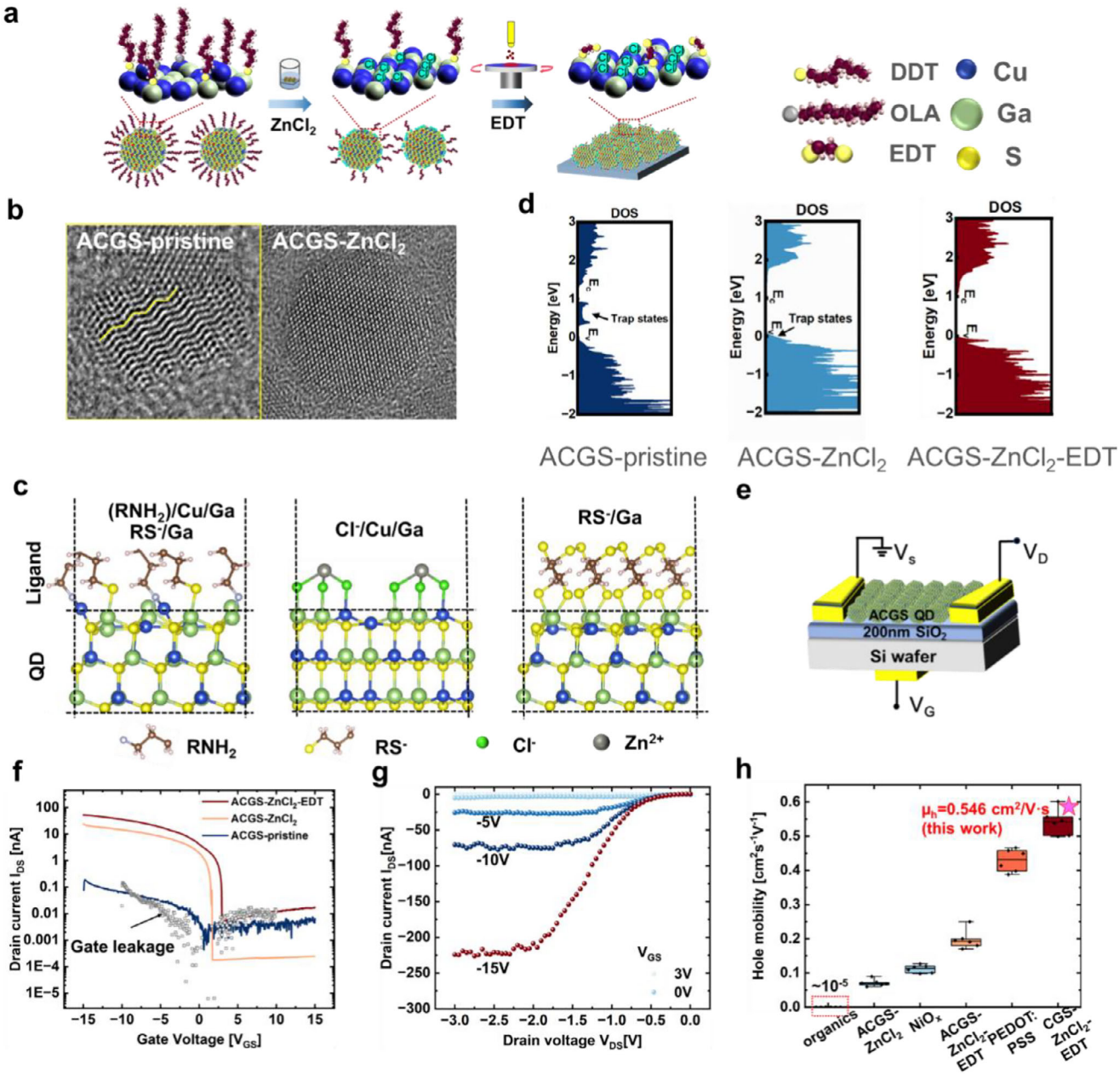

FIGURE 2 | Hybrid ligand exchange process and improved hole transport. (a) Schematic Illustration of hybrid ligand exchange with $ZnCl_2$ and EDT of QDs. (b) High-resolution TEM images of ACGS QDs before (ACGS-pristine) and after $ZnCl_2$ (ACGS-$ZnCl_2$) ligand exchange; the lattice distortions due to steric hindrance induced by bulky organic ligand are highlighted in yellow solid lines. (c) Relaxed configurations of (112) ACGS surface with ligands of long-chain organic ligand (RNH_2, RS^-), Cl^- ions, and short-thiolates (RS^-). (d) Calculated density of states (DOS) for long-chain ligand-passivated (ACGS-pristine) and short-ligand passivated (ACGS-$ZnCl_2$, ACGS-$ZnCl_2$-EDT) QD surface. (e) Device structure of QD-TFT based on ACGS QD solids channel on a silicon substrate. (f) Transfer characteristics of QD-TFT based on ACGS-pristine, ACGS-$ZnCl_2$, ACGS-$ZnCl_2$-EDT QD solids as transport channels. The grey open squares indicate gate leakage current. (g) Output characteristics of ACGS-$ZnCl_2$-EDT QDs with V_{GS} swept from −15 to 3 V. (h) Calculated hole mobility in the linear region for organic hole functional material: PVK, TFB, PEDOT:PSS, as well as inorganic material: NiO_x, CGS-$ZnCl_2$-EDT, ACGS-$ZnCl_2$, ACGS-$ZnCl_2$-EDT. The highest mobility achieved is denoted with a red star for CGS- $ZnCl_2$-EDT prepared in this work. The hole mobilities of TFB and PVK are both in the range of 10^{-5} cm^2 V^{-1} s^{-1}.

as reduced shallow, acceptor-like V_{Cu} (copper vacancy) defects, therefore decreasing the hole current.

The corresponding hole mobility (μ_h) in the linear working region was calculated based on the transfer characteristics (see Experiment Section). For comparison, the transfer characteristics of PEDOT:PSS, NiO_x, TFB, and PVK were also measured and summarized (Figure S14d; Figure 2h). The calculated μ_h in ACGS (AAC = 10%) and CGS QD solids could reach 0.191 cm^2 V^{-1} s^1 and 0.546 cm^2 V^{-1} s^1, respectively, compared to that of 10^{-5} for organic HTL and 0.127 cm^2V^{-1} s^1 for NiO_x thin-films. With μ_h calculated, the hole concentration (p_h) was predicted through current-voltage (*J*–*V*) characteristics of the two-terminal device structure: glass/ITO/ACGS QD/MoO_3/Ag (Figure S14e).

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Table S5 summarizes μ_h and p_h for chalcopyrite QD solids after hybrid ligand exchange prepared in this work. ACGS QDs exhibit nominal hole mobility but relatively lower hole concentration than PVK.

To evaluate the carrier mobility in the operating LED devices, we also performed space charge limited current (SCLC) measurement on a single-carrier device. According to the Mott–Gurney model, as voltage increases, the following equation applies:

$$J = \frac{9\mu\varepsilon_o\varepsilon_r V^2}{8l^3} \quad (1)$$

where ε_o and ε_r are the dielectric constants of air and the relative dielectric constant of quasi-2D perovskite, which can be measured from *C-V* analysis (see Experiment section for more details). L represents the thickness of the perovskite films, which is 20 nm. Thus, the carrier mobility, μ, could be extracted from the slope of linearly-fitted *log J-log V* characteristics (Figure S15). The estimated electron mobility was calculated to be 2.34×10^{-2} $cm^2\ V^{-1}\ s^1$, the hole mobility was 6.61×10^{-3} for the ACGS QD solids-based device, ten times higher than that estimated for PVK (4.19×10^{-4} $cm^2\ V^{-1}\ s^1$). Both QD-TFT measurement and SCLC calculations confirm the improved carrier mobility of ACGS QD solids, compared to traditional organic HTL materials.

2.3 | Suppressed Halide Migration and Defect Passivation in Quasi-2D Perovskites

Based on previous studies on interface engineering in blue PeLEDs, the physico-chemistry of HTL has demonstrated enormous effects on crystallization kinetics and interface defect formation of quasi-2D perovskite [42, 43]. Likewise, in this work, it is expected that the surface chemistry of ACGS QDs should be correlated with ion migration and subsequent defect formation in quasi-2D perovskite. From XPS measurement and DFT calculations in Figure 2c, the electronegative Lewis base Cl^- as X-type ligands provides covalent bond formation with electrophilic surface sites of ACGS QDs (Cu^+, Ga^{3+}), leaving densely-packed Zn^{2+} on the outer surface of QDs. Conceptually, Zn^{2+} capped on the ACGS surface could bind to halides (Br^-/Cl^-) in quasi-2D perovskites through electrostatic attraction, which may be beneficial to suppress halide vacancy defect formation and phase separation [42, 44, 45]. In this session, the influences of ACGS QDs on the exciton recombination lifetime of quasi-2D perovskite were investigated through time-resolved photoluminescence (TRPL) and XPS. Excitons trapped by vacancy defect states or uncoordinated Pb^{2+} should be subject to non-radiative recombination, which will be reflected in carrier lifetime [15, 46].

Figure 3 displays high-resolution XPS emission of Pb 4*f*, Br 3*d*, and Cl 2*p* photoemissions of quasi-2D perovskite deposited on ACGS QD solids. The quasi-2D perovskite layer deposited on PEDOT:PSS substrate is indicated as the "control" sample, compared to that deposited on ACGS-$ZnCl_2$-EDT QD solids. From Figure 3b,c, a 0.3 eV shift to higher binding energies in photoemission peaks of Cl $2p_{1/2}$ (199.48 eV) and Cl $2p_{3/2}$ (197.88 eV), as well as Br $3d_{1/2}$ (69.26 eV). This shift can be attributed to the strong electrostatic interaction between Zn^{2+} and halide anions. The Zn^{2+} cations attached to the surface of ACGS QDs were detected to diffuse into the perovskite during the annealing process (Figure S16). In addition, from Figure 3a, the core-level peaks of Pb 4*f* shift to higher binding energies, due to increased electron-affinity of Cl^-, suggesting a more stable octahedral $[PbX_6]^{4-}$ structure. Therefore, it is concluded that the incorporation of ACGS QD solids facilitates passivating halide dangling bonds and stabilizing the crystal structure through the formation of Zn—Br/Cl bond, leading to reduced interface defects. Consequently, the stabilization of halide interstitials via Zn^{2+} incorporation could alleviate vacancy defect formation, which tends to provide major ionic migration pathways in mixed-halide perovskite [47]. Similar observations have been reported for alkaline salts, Rb^+ and K^+, to suppress the detrimental effects of chlorine in photo-induced phase separation [42, 48, 49]. Due to suppressed interfacial exciton quenching and better defect passivation, the overall steady-state photoluminescence (PL) intensity of quasi-2D perovskite films crystalized on ACGS-$ZnCl_2$-EDT QD solids is significantly improved compared to those on ACGS-pristine, as well as PEDOT:PSS substrate (Figure 3d).

The charge carrier dynamics at the perovskite/HTL heterojunction sites and within the quasi-2D perovskite were examined by time-resolved PL (TRPL) analysis, as shown in Figure 3e. We extracted an average recombination lifetime of 6.46, 1.26, and 1.76 ns for quasi-2D perovskite deposited on PEDOT:PSS (control), PEDOT:PSS/ACGS-pristine, and PEDOT:PSS/ACGS-$ZnCl_2$-EDT QD solids, respectively. The extracted PL decay lifetimes (τ_1, τ_2) and their proportionality coefficients (A_1, A_2) are presented in Table S6 using second-order exponential functions (see Experimental Section). In a typical quasi-2D perovskite, the fast decay process (τ_1) is usually considered to be the lifetime of excitonic emission, while the slow decay process (τ_2) is strongly correlated with charge transfer between PEDOT:PSS and perovskite [50]. The proportionality coefficient of the fast-decay component (A_1), increases to 0.81 for perovskite deposited on ACGS-$ZnCl_2$-EDT substrate, compared to 0.50 and 0.75 for that deposited on PEDOT:PSS, and ACGS-pristine substrate, respectively. This result indicates strengthened radiative recombination in the quasi-2D perovskite layer. In addition, τ_2 fitted for perovskite deposited on ACGS-$ZnCl_2$-EDT QD solids substrate (3.98 ns) is reduced compared to that on PEDOT:PSS (7.16 ns), reflecting faster hole transport into ACGS-$ZnCl_2$-EDT QD solids for photogenerated carriers. We also performed TRPL analysis on quartz glass substrate to exclude the effects of charge transport (Figure S17 and Table S7). As expected, the overall lifetime is shortened from 9.19 ns (quartz glass) to 1.54 ns (ACGS-pristine) and 4.13 ns (ACGS-$ZnCl_2$-EDT), due to effective charge extraction of ACGS QD solids. Enhanced proportionality of fast-decay component (A_1), as well as shortened slow-decay lifetime (τ_2) for perovskite deposited on ACGS-$ZnCl_2$-EDT QD solids substrate, suggests accelerated exciton radiative recombination due to passivated lattice defect in quasi-2D perovskites on ACGS-$ZnCl_2$-EDT QD solids. We also observed significantly enlarged crystal dimensions of quasi-2D perovskite deposited on ACGS-$ZnCl_2$-EDT QD solids in Figure 3f, which could be attributed to increased hydrophilic properties of ACGS-$ZnCl_2$-EDT QD solids compared to PVK (Figure S18), leading to homogeneous film coverage and a uniform nucleation process. The reduced grain boundaries also contributed to suppressed ionic migration and improved structural stabilities.

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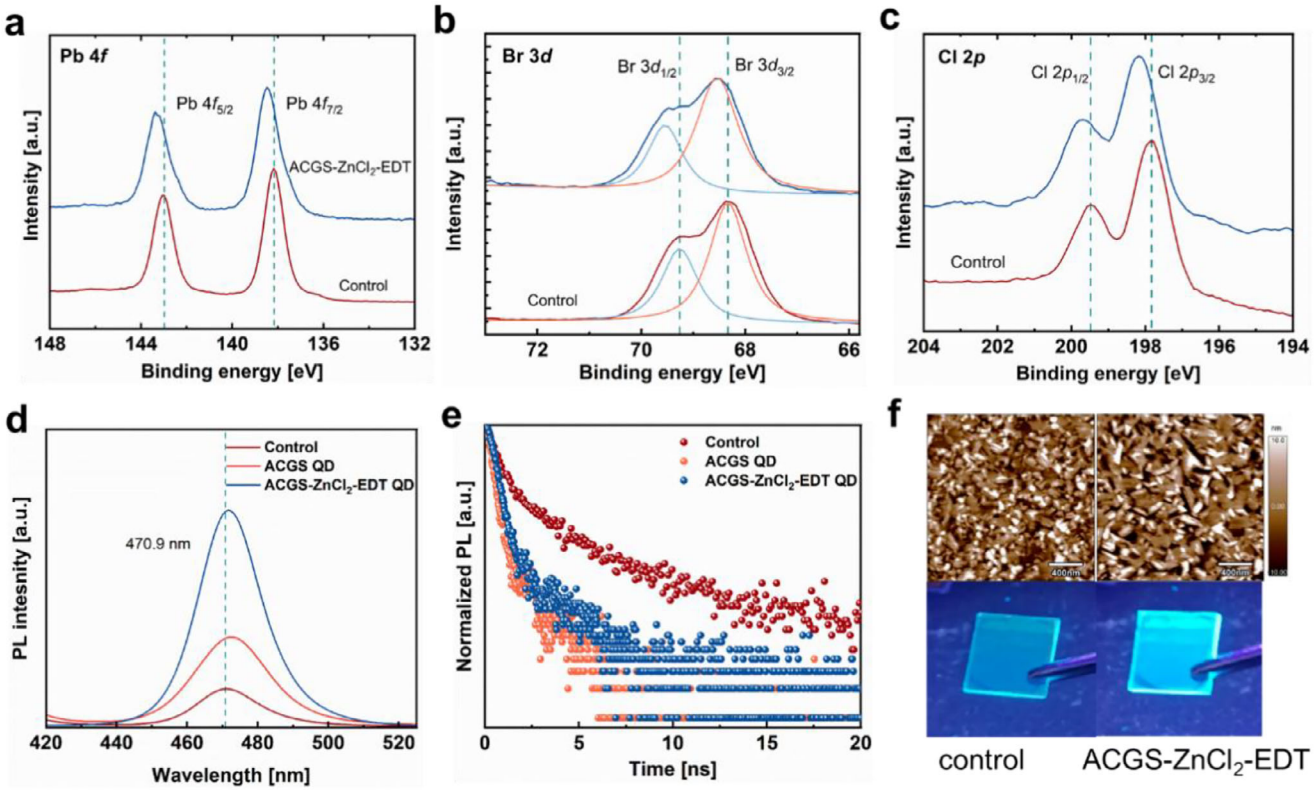

FIGURE 3 | Influences of ACGS QD solids on chemical bonding states and charge carrier dynamics in quasi-2D perovskite. High-resolution XPS spectra of the perovskite films deposited on PEDOT:PSS (control) and ACGS-$ZnCl_2$-EDT substrate for: (a) Pb 4*f*, (b) Br 3*d*, and (c) Cl 2*p*. (d) Steady-state PL spectra and (e) TRPL decay of perovskite films deposited on PEDOT:PSS (control), ACGS QD solids, and ACGS-$ZnCl_2$-EDT QD solids substrate. (f) Surface morphology (top) and picture (bottom) of quasi-2D perovskite films excited under UV excitation of 365 nm.

To elucidate the suppressed ion migration in quasi-2D perovskites due to Zn^{2+} incorporation, qualitative activation energy (E_A) for ion migration was obtained through thermal conductance (σ(T)) measurement for lateral devices: Au/perovskites/Au. E_A was evaluated using the Nernst–Einstein equation:

$$\sigma(T) = \sigma_o T \exp(-E_A/k_B T) \quad (2)$$

where k_B is the Boltzmann constant, and σ_o is a constant; E_A can be obtained by calculating the slope of *ln (σT)-1/kT* [51]. The fabrication details and original *J–V* characteristics are shown in Figure S19. For devices based on PVK as HTL (deivce A), two distinct linear regions could be identified. The first linear region (from 260 to 290 K) shows an E_A of 15.09 ± 3 meV; the second linear region (from 290 to 320 K) shows a much larger E_A of 242.2 ±24.7 meV. In the low-temperature region, the ion migration was inhibited; thus, the smaller E_A can be assigned to charge carrier transport in quasi-2D perovskite. Compared to electron conduction, electro-migration dominates the conductivity at relatively high temperature (310–350 K) due to its large activation energy. Thus, the activation energy of halide ion migration (Cl^-) in quasi-2D perovskite deposited on PVK is determined to be 0.242 eV [52]. While for device B, based on ACGS QD solids as HTL, the ion conduction is almost negligible from temperature rises up to 340 K, and a derived small activation energy of 0.0159 eV is attributed to electronic conduction. Furthermore, the ion conductivity of device B is much lower than that of device A. And the zero-point deviation occurred in device B (~10–20nA), which is much smaller than that for device A (~100nA), indicating suppressed hysteresis of ionic movement. All these discussions indicate that the ion migration in the quasi-2D perovskite films deposited on ACGS QD solids is significantly suppressed due to the ionic coordination bond between Zn^{2+} and Cl^-.

The crystallization regulation effects of ACGS QD solids on quasi-2D perovskite were also evaluated from XRD and UV–vis absorption spectra. Figure S20a showed distinct absorption peaks attributed to the small-n (n = 1, 2, 3) phases. XRD diffraction pattern shows the diffraction peaks at 15.6° and 31.3° corresponding to (100) and (200) planes of 3D $CsPbBr_3$ perovskites. While, diffraction peaks at 12.8° and 27.8° indicate layered quasi-2D phases (Figure S20b). The negligible influences of ACGS QD solids on the phase distribution of quasi-2D perovskite substantiate that the improved photoluminescence and exciton recombination primarily come from passivated halide-related lattice defects with decorated ACGS QD surface and enhanced hole transport.

2.4 | Defect Physics and Highly-Efficient, Stable Blue PeLEDs

To corroborate advantageous effects of ACGS QD solids on carrier transport and defect passivation in PeLEDs based on quasi-2D perovskite, blue PeLEDs devices were prepared with the structure: glass/ indium tin oxide

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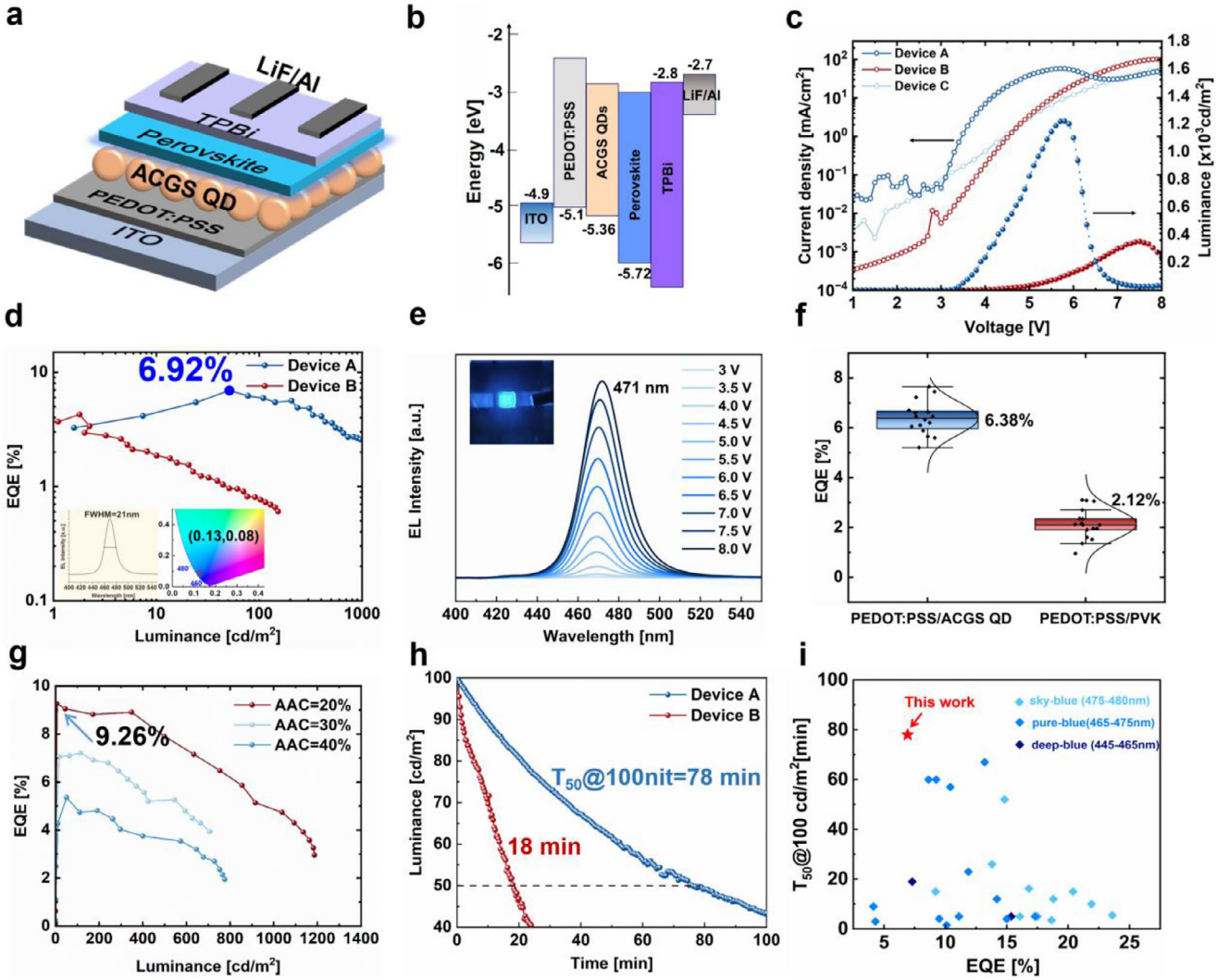

FIGURE 4 | Performance analysis of blue perovskite light-emitting diodes. (a) Illustrations of PeLED device architecture based on ACGS QD as hole transport layer developed in this work (Device A), and (b) corresponding schematic band diagram. (c) Current density-luminescence-voltage (*J-L-V*) characteristics of PeLEDs based on PEDOT:PSS/ ACGS QD (Device A), PEDOT:PSS/ PVK (Device B), and ACGS QD as hole injection layer (Device C), and (d) corresponding external quantum efficiency-luminance (*EQE-L*) characteristics of PeLEDs. (e) EL spectrum of champion PeLEDs operating at elevated voltages. (f) Histogram of maximum EQE measured for 20 individual devices. (g) *EQE-L* characteristics of device A based on ACGS QD solids synthesized with a varied AAC ratio in the precursor. (h) Operating lifetime decay curves luminance-time (*L-t*) of device A and device B based on PVK as HIL. (i) Summarized operational lifetime (normalized to T_{50}@100 cd/m^2) against device EQE of quasi-2D blue PeLED, categorized into deep-blue, pure-blue, and sky-blue according to the literature.

(ITO)/PEDOT:PSS/ACGS QD/perovskite/1,3,5-tris(1-phenyl-1H-benzimidazol-2-yl)benzene(TPBi)/LiF/Al (device A), glass/ ITO/PEDOT:PSS/PVK/perovskite/TPBi/LiF/Al (device B) and glass/ ITO/ACGS QD/PVK/perovskite/TPBi/LiF/Al (device C) in which ACGS QD dictates ACGS-$ZnCl_2$-EDT QD solids.

The device architecture (device A) was optimized based on the band landscape as schematically drawn in Figure 4a, and the corresponding energy band level of quasi-2D perovskite was obtained from UPS measurement (Figure 4b; Figure S21). The E_F and VBM of quasi-2D perovskite films prepared here are determined to be 4.92 and −5.72 eV, consistent with microscopic KPFM results (Figure S7). To match the VBM of perovskite, the ACGS QDs were delicately synthesized with an ACG ratio of 1:8 and a varied AAC ratio from 10% to 50%, of which the VBM would lie between −5.3 and −5.7 eV. Figure 4c,d demonstrate electroluminescent performances of devices A, B, and C. From Figure 4c, device C exhibit shunting diode and low EQE. Compared to device B, device A preserves rapidly-increased current density after reaching turn-on voltage (3.2 V), due to increased hole conductivities of ACGS QD solids, preventing non-radiative recombination losses, thus maximizing exciton energy transfer, leading to a maximum EQE and luminance of 6.92%, and 1222 cd/m^2, respectively. Figure 4e shows the EL spectrum of champion blue PeLEDs based on ACGS QDs prepared in this work. The EL emission peak at 471 nm with a narrow emission bandwidth of 21 nm matches well the PL spectrum (Figure 3d). The spectral stabilities of the EL emission plot strongly support the prominent advantages in preventing deep-level trap formation in quasi-2D perovskite based on ACGS QD solids. A histogram of the maximum EQE values for 20 devices (Figure 4f) shows an average EQE of 6.38%, compared to that of 2.12% for

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device B with PVK as HTL. Figure 4g and Figure S22 display the device performances of device A with a varied AAC ratio of ACGS QD since the applied AAC ratio during QD synthesis is of vital importance in controlling hole mobility and hole transport barrier of QD solids. Table S8 summarizes the detailed device performances. As AAC increases to 20%, the maximum EQE could reach 9.26%. Further doping with Ag tends to degrade device performances due to dramatically dropped hole mobility of HTL. The measured operational lifetime when the luminance dropped to half of the initial luminance at 100 cd/m^2 (T_{50}@100nit) of device A is significantly extended to 78 min, which is more than four times longer than device B (18 min) (Figure 4h). The enhanced stabilities indicate optimal charge management, thus preserving a high excitonic recombination rate under long-term operation of PeLEDs. To highlight the stability enhancement of PeLEDs due to ACGS QDs, we also prepared NiO_x-based PeLEDs (device D), with the structure of glass/ indium tin oxide (ITO)/NiO_x/ACGS QDs/perovskite/TPBi/LiF/Al (Figure S23). The highest EQE achieved for device D is 5.98%, but the lifetime is rather low, which might be due to the weak attachment of QD to NiO nanoparticles [22]. Figure 4i summarizes the reported EQE against lifetime (normalized to T_{50}@100 cd/m^2) for PeLEDs based on quasi-2D perovskite, and the corresponding literature can be found in Table S9. The significantly improved operational lifetime of pure-blue quasi-2D PeLEDs prepared in this work, manifested as world-class stable devices.

Built on previous studies on degradation mechanisms of PeLEDs with quasi-2D perovskite, we are convinced that concentrated deep-level defects will arise under luminescent operation either at the thin film interface or inside the perovskite layer [23, 26, 53]. To dive deep into defect physics in quasi-2D perovskite and blue PeLEDs on introducing ACGS QD solids as HTL, we aim to quantitatively validate defect properties from device-level electrical diode analysis. Specifically, thermal admittance spectroscopy (TAS) and capacitance-profiling (*C-V*) were conducted on LED devices and hole-only devices (HOD) based on the architecture of devices A and B. From raw capacitance-frequency (*C-f*) characteristics provided in Figure S24, the activation energy (E_A) of the deep-level defect could be extracted, which is 0.41 eV (device A) and 0.60 eV (device B), respectively (Figure 5a). Moreover, the calculated density of states (DOS) for these deep-level traps is 5.42×10^{16} cm^{-3}·eV^{-1} (device A), and 6.16×10^{17} cm^{-3}·eV^{-1} (device B), respectively (Figure 5b). Figure S25 plots the DOS distribution, from which an energetically-narrow defect distribution and reduced maximum defect density can be revealed for device A. Notably, TAS were sensitive to energetically-discrete and spatially-distributed defects, and only the defect in the space charge region could contribute to junction capacitance [54]. For perovskite-based thin-film devices with a typical p-i-n structure, the perovskite layer is almost fully-depleted [55]. These results suggest that in device A, the deep-level traps are substantially weakened throughout the emissive layer. Though the physical/chemical origin of these defects remains further investigation, the trap-assisted non-radiative recombination is surely to be suppressed in device A. The detailed calculations on E_A and DOS are provided in the experiment sections. Compiling the above solid discussions on defect physics and chemical bonding analysis, it is thus ascertained that ACGS QD solids as HTL provide effective passivation of deep-level defects (i.e., chlorine vacancy) in quasi-2D perovskite through functionalized surface ligand. The enhanced hole injection firmly guarantees excitonic recombination while circumventing charge accumulation at the HTL/perovskite interface; as a result, both bulk and interfacial non-radiative recombination can be sufficiently impeded during the LED operation.

The charge injection barrier was systematically analyzed from *C-V* profiling of hole-only devices (HOD), as shown in Figure 5c, and the corresponding Mott–Schottky (MS) analysis was provided in Figure S26a. The built-in potential (V_{bi}) of HOD based on ACGS QD solids is 0.21 V, profoundly lower than that based on PVK (0.75 V), due to the higher WF level of ACGS QDs, which helps to reduce Fermi-level splitting with perovskite (Figure 5d). Thus, ACGS QDs facilitate an overall reduced band-bending and mitigated hole transport barrier. Consistently, the capacitance value of HOD based on ACGS QDs (device A) rises dramatically to 3.40 nF compared to that with PVK (device B), which stayed almost constant around 1.75 nF until the bias of 3 V, featuring effective charge injection from ACGS QD solids to perovskite [56]. Similar results were also determined in *C-V* measurement for PeLED devices, certifying lower V_{bi} for device A (1.8 V) compared to device B (3.0 V) (Figure S26). Consequently, the total current collected from HOD based on ACGS QD (device A) is 10 fold higher than that based on PVK (device B).

3 | Discussion

In summary, we have investigated chalcopyrite ACGS QDs as a potential inorganic hole transport material to achieve highly-efficient, stable blue PeLEDs. Engineering ACGS QDs by size, composition, and surface chemistry has realized high-mobility QD solids and lowered hole injection barrier, allowing for enhanced hole transport and balanced exciton formation in quasi-2D perovskite. The dual-functionalized surface ligand groups of ACGS QD also behave as "defect passivator" anchoring to halide sites, leading to fewer trap states and stabilized crystal structure of quasi-2D perovskite. With balanced charge transport and suppressed ion migration in quasi-2D perovskites through ACGS QDs incorporation, the device lifetime is considerably prolonged to the hour scale. This work provides a creative strategy to replace traditional vacuum-deposited, high-quality inorganic semiconductor thin films with facile, solution-processable QD solids, with identical electronic properties, thereby allowing us to harness new combinations of optical, electrical, and physical properties of QD solids, delivering numerous advantages that cannot be easily-achieved through a single inorganic–organic material system.

Most importantly, we bring about interfacial modifications via surface chemistry engineering of QD through chemically-interacting with the perovskite crystal, offering a new strategy to suppress defect-assisted recombination losses in quasi-2D perovskite, besides solving complicated crystallization pathways and phase control issues. It is feasible that inorganic chalcopyrite QD can be engineered purposely toward composition, size, shape, and surface chemistry as a promising "interfacial defect passivator" with perovskite, supporting enhanced interfacial carrier

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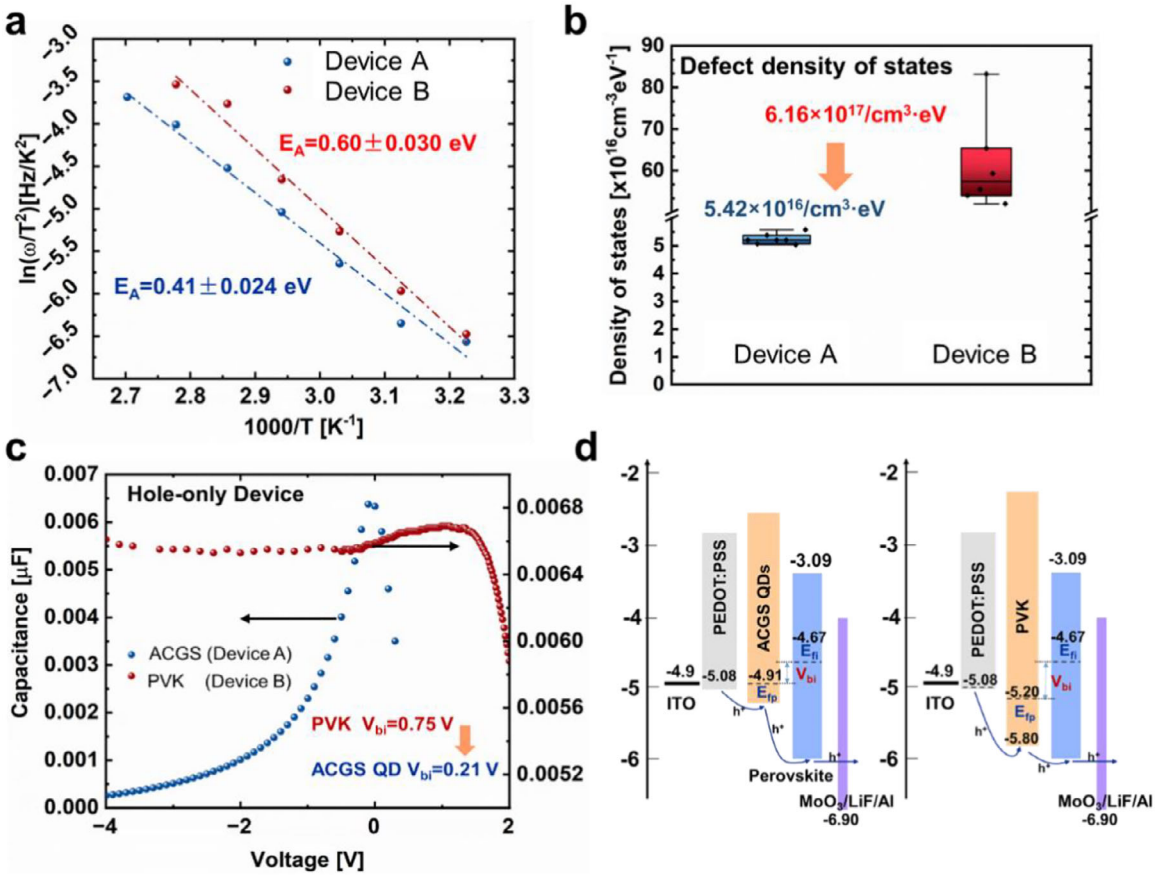

FIGURE 5 | Defect physics and diode analysis of PeLEDs. (a) Arrhenius plot of critical frequency to extract activation energy of deep-level defects for device A and device B. (b) Comparisons of maximum density of states (DOS) of deep-level defects in PeLEDs for device A and device B. (c) Capacitance-voltage (*C-V*) characteristics of hole-only devices (HOD) based on ACGS QD (device A), and PVK (device B). The corresponding built-in potential (V_{bi}) extracted from the Mott–Schottky plot is also highlighted. (d) Band diagram illustration of V_{bi} in HOD. The WF position is derived from the KPFM measurement and indicated as dashed blue lines. E_{fp} and E_{fi} represent the quasi-fermi level for HTL and the quasi-2D perovskite, respectively. Due to high hole concentration of p-type ACGS, the WF is elevated to −4.91 eV for ACGS QD solids, reducing the Fermi-level splitting from 530 to 240 mV, thus alleviating the band bending of HTL/perovskite hetero-junction significantly and enhancing hole transport from HTL to perovskite.

extraction and transport. Aiming for regulated crystallization and phase modulation for quasi-2D blue perovskite, surface ligand exchange with benzene ring, phosphate group (-P = O-), and other regularly-used organic amines on ACGS QDs will be focused on in future work for further enhancement in device efficiency and lifetime.

4 | Experiment Sections

4.1 | Materials

Oleylamine (OLA, 80%–90%), 1-dodecanethiol (DDT, 98%), octane (>99% GC), cyclo-hexane(>99% GC), gallium acetylacetonate (99.99% metal basis), gallium chloride ($Ga(acac)_3$, 99.999% metal basis), cuprous chloride (metal basis 99.95%), cuprous iodide (metal basis 99.95%), sublimed sulfur (AR, ≥99.5%(T)), ethanedithiol (97%) were purchased from aladdin. Zinc chloride ($ZnCl_2$) was purchased from Alfa Aesar (Ultra dry, 99.999% metal basis). Silver iodide (premion metal basis 99.999%) was purchased from Thermo Scientific. CsBr (99.999%, metal basis), $PbBr_2$ (99.999%, metal basis), and $PbCl_2$ (99.999%, metal basis) were purchased from Alfa Aesar. DMSO (99.9%) was purchased from Sigma–Aldrich. PEABr were purchased from Xi'an Polymer Light Technology Corp. TPBi, and LiF were purchased from Luminescence Technology Corp. NiO nanoparticles were purchased from Liao'ning You Xuan Renewable Energy Technology Corp.

4.2 | Synthesis of Chalcopyrite QD

In a typical synthesis of ACGS QD for PeLED fabrications, 0.025 mmol of AgI, 0.225 mmol of CuI, 1 mmol $Ga(acac)_3$, 10 mL OLA, 10 mL ODE, and 2 mL DDT were placed in a 50 mL three-neck flask by stirring. The mixture was heated to 100°C after degassing for 30 min. The temperature was then further increased to 180°C under an Argon (Ar) flow. At 140°C, a sulfur stock, which was obtained by ultrasonically dissolving 2 mmol sulfur in OLA, was rapidly injected into the mixture to initiate the reaction. During the injection, careful pressure control under N_2 flow was required to stabilize the temperature. The growth of QDs was maintained at 180°C and lasted for 5 min. The ACGS QDs were isolated by centrifugation using ethanol and dispersed in cyclohexane for $ZnCl_2$ ligand exchange.

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4.3 | Hybrid Ligand Exchange

The molecular weight of ACGS QDs was estimated as:

$$M = N_A \times \rho V_{QD}$$

where N_A denotes Avogadro's constant, which is 6.02×10^{23}. ρ is the density of chalcopyrite in CGS, which is 4.35 g/cm^3. Assuming the QD is spherical, and the diameter is determined from TEM measurement, which is 4.41 nm. Then the volume of the QD is calculated as:

$$V_{QD} = \frac{4}{3}\pi r^3$$

The calculated molecular weight of ACGS QDs is 40 kDa. The as-synthesized ACGS QDs were separated by centrifugation using ethanol and re-dispersed in 2 mL cyclohexane. The concentration of as-synthesized QDs was determined by evacuating the cyclohexane solvent from a 100 µL aliquot. For a single reaction using the aforementioned precursor amounts, the concentration typically ranged from 100 to 120 mg/mL. Then, 0.1 mmol (20 µL of 5 M $ZnCl_2$ in ethanol solution) was added to 400 µL ACGS QD dispersion (containing approximately 40 mg of QDs, or ~1 umol). The mixture was heated to 60°C and maintained at this temperature for 30 min. This corresponds toa nominal ligand-to-QD molar ratio of 100:1 (0.1 mmol:1 µmol). In practice, due to the large amount of organic ligand attached onto the surface of QDs, the actual amount of inorganic core is lower than the measured total mass. This results in a higher actual ligand-to-QD ratio than the calculated value. Nevertheless, estimating this ratio is still meaningful to optimize the ligand exchange process. The chlorine-treated QDs were separated by centrifuging and redispersed in octane. All processes were carried out in a fume hood under ambient conditions.

Solid-phase ligand exchange with EDT was conducted using a layer-by-layer approach in the N_2-filled glove box. A solution of EDT in acetonitrile (1% by volume) was applied onto the ACGS QD film and left there for 60 s. Subsequently, the film was spun at 1500 rpm for 40 s. To remove excess EDT and displaced DDT ligand, two consecutive rinsing steps were performed with neat acetonitrile via drop-casting, and the film was baked at 60°C for 30 min.

4.4 | Perovskite Precursor Preparations

The stoichiometric ratios of CsBr, $PbBr_2$, $PbCl_2$, and PEABr (molar ratio of CsBr:$PbBr_2$:$PbCl_2$:PEABr = 0.2:0.1:0.1:0.16) were dissolved in dimethyl sulfoxide (DMSO) with a concentration of 0.2 M and stirred overnight in a nitrogen glove box at 60°C to prepare the mixed precursor. The precursor solution is filtered through a 0.22 µm PTFE membrane prior to use.

4.5 | Device Fabrications

For PeLED device fabrications, the patterned ITO substrates were ultrasonically treated with detergent, deionized water, acetone, and isopropyl alcohol successively for 15 min. The surface of the ITO substrate was treated with oxygen plasma to improve its hydrophilicity. The PEDOT:PSS solution was spin-coated onto the ITO substrate at 4000 rpm and then annealed at 150°C for 20 min. After baking, the substrate was transferred to a nitrogen-filled glovebox for the deposition of the hole transport layer. For ACGS QD-based devices, ACGS-$ZnCl_2$ (ligand-to-QD ratio = 100:1) in octane (concentration:4 mg/mL) was first spin-coated at 3000 rpm for 30 s (thickness ~5 nm). Then a film-washing step was performed using EDT in acetonitrile (1 vol.%). The resulting ACGS-$ZnCl_2$-EDT QD thin-film was annealed at 60°C for 30 min. For PVK-based devices, PVK in chlorobenzene (concentration: 8 mg/mL) was spin-coated onto the PEDOT:PSS at 4000 rpm for 40 s, followed by baking at 120°C for 20 min (thickness ~25 nm). For the NiO_x-based device, NiO nanoparticles were dissolved in deionized water with a concentration of 10 mg/mL. Then the solution was spun onto the ITO substrate at 3000 rpm for 30 s, followed by annealing at 150°C for 15 min. Perovskite films were spin-coated from a precursor solution at 3000 rpm. After baking at 50°C for 5 min, the samples were transferred to a vacuum chamber ($< 5 \times 10^{-4}$ Pa) for the successive thermal evaporation of TPBi (40 nm), LiF (1 nm), and Al electrodes (100 nm). The active area of the device (4 mm^2) was defined by the overlap of the electrodes using a shadow mask during cathode deposition. Except for PEDOT:PSS and NiO_x film deposition, other processes were conducted in a N_2-filled glovebox.

The EL spectra characteristics of the PeLEDs were measured using a Shenzhen Pynect integrated sphere (50 mm in diameter) coupled with a Keithley 2400 source meter. The operation lifetime tests of PeLEDs were conducted using a ZJZCL-1 OLED aging lifespan test instrument. The *I–V* characteristics of the PeLEDs were collected on a system consisting of a PR-670 Spectra Scan Spectroradiometer coupled with a Keithley 2400 source meter.

The electron-only devices were prepared with the structure ITO (150 nm)/ZnMgO(20 nm)/perovskite/TPBi (40 nm)/LiF (1 nm)/Al (100 nm). The hole-only devices were prepared with the structure lTO (150 nm)/PEDOT:PSS (30 nm)/HTL/perovskite/MoO_3(10 nm)/LIF (1 nm)/Ag (100 nm). The deposition process of single-carrier devices was exactly the same as that described for LED devices.

4.6 | Characterizations

The absorption characteristics of QDs were measured with a UV–vis spectrometer (Model: UV2600, Shimadzu). The QD solution was diluted about 100 times using octane to reach an optical density of 2 at 300 nm. Crystal structures were analyzed by X-ray diffraction (XRD) (Model: Smartlab SE, Rigaku). Steady-state photoluminescence (PL) and TRPL spectra were recorded on an Edinburgh FLS920 PL spectrometer at an excitation wavelength of 365 nm. TEM images were obtained on JEM-3200FS from JEOL, and the energy-dispersive X-ray (EDX) spectra were obtained using X-max N 100 LTE spectrometer.

The film morphology was obtained by scanning electron microscopy (SEM), measured on (Model: Sigma 300, ZEISS) with a spectrometer (Xplore 30), inlens and secondary electron detector. The accelerating voltage is 0.02–30 kV with a step of 10 V. Atomic force microscopy (AFM) (Cypher ES) was employed

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for microscopic surface morphology, and the KPFM module was employed for surface potential mapping measurement. The sample chamber is specifically protected by N_2 to prevent degradation of perovskites during measurement. The work function (WF) is calculated using the equation:

$$\varphi_{sample} - \varphi_{tip} = eV_{CPD}$$

where φ_{sample}, φ_{tip}, and V_{CPD} denote the WF of the sample, the tip, and the average contact potential difference. φ_{tip} (AC 24O TM) is 4.9 eV used in this work.

The Ultraviolet photoelectron spectroscopy (UPS) was measured on ESCALAB XI+ from Thermo Fisher Scientific. He (I) UV radiation source (21.22 eV) from a He discharge lamp was used in the UPS measurements. X-ray photoelectron spectroscopy (XPS) was measured on Nexsa from Thermo Fisher Scientific. Al Kα lines were used as the excitation source, and the operation voltage is 12 kV. A survey scan was performed with a range of 100 eV, and a step of 1 eV, and a fine scan was performed with a step of 0.05 eV. The binding energy was calibrated with C 1*s* emission at 284.8 eV.

FTIR was performed by a Bruker spectrometer with an ATR hemisphere. The transmission mode was used, and the background scan is vaccum atmosphere without samples. TGA was performed from 20°C to 600°C at a heating rate of 10°C min under N2 using a Trios V3.2 system (Model: TGA55, TA Instruments). The hydrodynamic particle sizes of the QDs in solutions were determined using a particle-size analyser (Model: Nano ZS90, Malvern Zetasizer). The actual content of elements in QD was detected using ICP-OES (Model: 720ES (OES), Agilent). The QD solution was dried under nitrogen, and the obtained solid was dissolved in 3% nitric acid and deionized water for ICP-OES, respectively. The sample solution was diluted 10 000 times for ICP-OES measurements. Contact angle measurements were obtained on a Lauda Scientific instrument using deionized water (Model: LSA100).

The EL spectra characteristics of the PeLEDs were measured using a Shenzhen Pynect integrated sphere (50 mm in diameter) coupled with a Keithley 2400 sourcemeter. The operation lifetime tests of PeLEDs were conducted on a ZJZCL-1 OLED aging lifespan test instrument. The *I–V* characteristics of the PeLEDs were collected on a system comprised of a PR-670 Spectra Scan Spectroradiometer coupled with a Keithley 2400 sourcemeter.

4.7 | DFT Simulation Details

The passivated surface structures were simulated by constructing a slab model with the coupling of the (112) $CuGaS_2$ surface and adsorbed molecules. The slab was separated by a 15 Å vacuum space in the *z*-direction to avoid interaction between layers.

The structural optimization, total energy, and electronic states of these surface configurations were calculated by employing the Vienna ab initio simulation package (VASP) within the framework of density functional theory. In the calculations, the projector augmented wave method and the exchange-correlation functional of generalized gradient approximation (GGA) were adopted. The energy cut-off was set as 350 eV for the plane-wave basis-set expansion. The *k*-point of the Brillouin zone was 0.03 $Å^{-1}$ interval distribution of Monkhorst–Pack for the optimization of structures, and the *k*-point interval of the total energy self-consistent calculation was 0.02 $Å^{-1}$ or better. All the structures were fully optimized, and convergence thresholds were set as 10^{-5} eV in energy and 10^{-3} eV/Å in force.

Based on the calculated total energies, the ligand binding energy was estimated as:

$$E_b = E_{(CuGaS2-ligand)} - E_{CuGaS2} - \mu_{ligand}$$

where $E_{(CuGaS2\text{-}ligand)}$ and E_{CuGaS2} are the total energy of the passivated and bare slab, respectively, and μ_{ligand} is the chemical potential of the ligand.

4.8 | Thin Film Transistor Measurement

The QD-TFT was fabricated as the following steps: (1) A low-resistance (resistivity: 0.001–0.005 Ω•cm) p-type Si wafer with an oxide layer of 200 nm was cleaned by the SC-1 standard process. (2) NR-1000P (Futurrex) was spin-coated at 4000 rpm for 40 s, followed by a softbake at 150°C for 60 s. (3) The sample was exposed under UV light (365 nm) for 30 s, and then developed in RD6 for 15 s. (4) Cr/Au (10 nm/30 nm) drain/source electrode was deposited through electron-beam evaporation, followed by a lift-off process. QDs solution dispersed (20 mg/mL) in octane was spin-coated onto the pre-patterned silicon substrate at 1500 rpm for 40 s, followed by baking at 60°C for 30 min. (5) The gate electrode was probed directly on the bottom silicon surface through mechanically-scratching thin film stacks on top.

The device yield measurement of QD transistors was performed using a manual probe station under an ambient environment. To avoid a short circuit, an insulated glass was placed between the wafer and the sample stage. For the measurement of transfer curves, the bottom gate was connected to an SMU of the Keithley 4200. One drain was connected to another SMU, the source was connected to the ground, and the bulk was left as none. It is important to mention that the measurement was conducted under a dark, relaxed state since the photo-response of CGS QDs is non-negligible.

The hole mobility (μ_h) in the linear region of working TFTs can be figured out by using the equation:

$$\mu_h = \frac{dI_{DS}}{dV_{GS}} \frac{l}{wC_{ox}V_{DS}} \tag{3}$$

where *l* and *w* are the channel length and width of QD-TFT, which are 50 μm and 50 μm, respectively. 200 nm-thick SiO_2 was deposited as a dielectric layer. Cox is calculated from the equation:

$$C_{ox} = \varepsilon_{ox}/t$$

where the dielectric constant (εox) is estimated to be 3.9 for SiO_2. *t* refer to the total thickness of insulating oxide. $\frac{dI_{DS}}{dV_{GS}}$ could be obtained from the slope of the linear approximation of the I_{DS}-V_{GS} plot at negative bias.

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The hole concentration (p_h) is estimated through the equation:

$$p_h = \frac{\sigma_h}{q\mu_h} \tag{4}$$

where σ_h, q, and μ_h denote the conductivity, electron charge, and hole mobility, respectively. σ_h could be obtained from J–V curves of two-terminal devices based on various HTL. Substituting the estimated value in Equation (4), p_h could be calculated.

4.9 | Thermal Admittance and Mott–Schottky Analysis

The capacitance spectra of perovskite solar cells were measured using an impedance analyzer from Tonghui Electronics (model: TH2838S) together with a home-built cryosystem temperature controller from LakeShore. Each capacitance spectrum was acquired at zero bias within the frequency range between 20 Hz and 1 M hz at a given constant temperature. To avoid probe-induced damage to the device, a thin indium (In) pad was placed underneath the probe, and then placed onto the Al electrode. The connected equivalent circuit model was used for capacitance calculations.

The maximum frequency (ω_0) that a defect at an energy level E_ω within the bandgap can contribute to the junction capacitance is written as:

$$\omega_0 = 2N_v\, \upsilon_{th}\sigma \exp\left(-\frac{E_\omega}{kT}\right) = \gamma_0\, T^2 \exp\left(-\frac{E_\omega}{kT}\right) \tag{5}$$

N_v and υ_{th} represent the density of state and thermal velocity of valence band holes, respectively. Their product can be generally written as $\gamma_0 T^2$. The temperature-independent property γ_0 is denoted as "attempt-to-escape" frequency. ω_0 can be derived by finding the maxima of "scaled derivative" ($-\omega/kT\ dC/d\omega$) as a function of the frequency axis. The activation energy level (E_A) of defects is estimated by drawing an Arrhenius plot between inflection frequency divided by T^2 (ω_0/T^2) vs. $1/T$ [26]. E_A is conceived as a conjunct effect of migration barrier and defect formation energy [33].

The electronic density of states (DOS) spectroscopy of a defect can be determined from "the scaled derivative":

$$N_t\ (E_\omega) = -\frac{V_{bi}}{qw}\frac{\omega}{kT}\frac{dC}{d\omega} \tag{6}$$

where V_{bi} and w represent the built-in electric field and depletion width of the device. V_{bi} is determined by the Mott–Schottky plot of capacitance-voltage profiling measurement, which is 1.8 and 3.0 V for the device with ACGS QD and PVK as HTL, respectively. The depletion region width at zero bias is determined from capacitance-voltage (C-V) characteristics for the LED device with the equation:

$$C = \frac{\varepsilon_o \varepsilon_r}{w} \tag{7}$$

where ε_o and ε_r denote the dielectric constant for vacuum and quasi-2D perovskite, respectively. ε_r is estimated using the distinct geometry capacitance (C_{geo}) in TAS measurement under high-frequency excitation. Based on the serial connection of HTL, perovskite, and ETL, and the high capacitance of HTL and ETL due to the thin layer, the geometry capacitance of the device is fully determined by the perovskite layer. Therefore, the C_{geo} is 1.53 and 1.60 nF for PeLEDs with PVK and ACGS QDs, respectively. Taking into account the thickness of the perovskite layer, which is roughly estimated to be within the range of 40–50 nm, the ε_r of quasi-2D perovskite is calculated to be 1.63–1.73, consistent with reported values [57]. Furthermore, the capacitance at zero bias is measured to be 1.80 and 2.11 nF for PeLEDs with PVK and ACGS QDs as HTL, respectively. Thus, the depletion region width for PeLEDs with PVK and ACGS QDs is calculated to be 30.09 and 25.69 nm, respectively. Substituting the $V_{bi,}$ w, and scaled derivative $-\omega/kT\ dC/d\omega$, the DOS could be achieved using Equation (6).

Author Contributions

Chunlei Yang and Xue Zheng conceived the idea and designed the experiment. Chunlei Yang, Wenjun Hou, and Yang Bai provided financial support, supervised the work, and oversaw the data acquisition. Yuanzhi Wang contributed to device optimizations, fabrications, and characterizations. Wenjie Ming was in charge of all morphology and electrical analysis using atomic force microscopy. Guohua Zhong aided in DFT simulations and analysis. Yan Shao conducted the photolithography techniques for the thin-film transistor, and Chuanlai Ren assisted with the corresponding thin-film transistor measurements. Xue Zheng and Yuanzhi Wang wrote the original draft of the manuscript. Jie Zhang, Ming Chen, and Haoran Wang helped organize the data and figures and provided meaningful suggestions. Chunlei Yang, Wenjun Hou, and Jian-Xin Tang provided major revisions. All authors discussed the results and commented on the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the plots within this paper and the other findings of this study are available from the corresponding authors upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

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